A Log-Linear IRT Accumulator Model of Early Word Learning

Draft specification (revised: 2PL extension)

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1 Research questions driving the modeling

The model is organized to answer four research questions. The baseline model is the minimal specification that lets each be posed and answered; each planned variant exists to sharpen one of them.

RQ1. How well do frequencies reconstruct word difficulties? Braginsky et al. (2019) showed frequency is only modestly related to AoA within lexical classes. We test this within our model by asking: how much of word-level residual difficulty ψ_j does $\log p_j$ explain?

RQ2. Does accumulation start near birth, or later? Mollica & Piantadosi (2017) estimated $s \approx 2$ months for comprehension. We re-examine s in the CDI production data, where we expect s to be larger but still informative.

RQ3. Is an acceleration term needed? Bergelson (2020) argued that older infants are "better learners" — they extract more from the same input. We test this by freeing δ, the age rate-change exponent; $\delta > 0$ would corroborate a comprehension-boost-like rate increase.

RQ4. How much of child-level outcome variance is input-driven? This is the σ_α question: once we pin the variance of CDS input rate r_i from external data, how much residual child variance must be absorbed by a learning-efficiency term α_i?

Each RQ becomes a posterior summary or a planned model comparison. The mapping from RQ to model structure is in Table **??**.

RQ	Quantity	How answered
1	$r(\hat{\psi}_j, \log p_j)$, $R^2_{\psi \sim \log p}$	Posterior in baseline; confirmed with free β, β_c variants
2	s	Posterior in baseline with prior $\mathcal{N}(4.5, 2^2)$ truncated
3	δ	Posterior in baseline with prior $\mathcal{N}(0, 0.5^2)$
4	$\pi_\alpha = \sigma_\alpha^2/(\sigma_r^2 + \sigma_\alpha^2)$	Derived posterior; σ_r^2 pinned from external data

Table 1: Mapping from research questions to model quantities.

2 One-paragraph summary

A child i of age a_i has, on average, heard word j approximately $T_{ij} = r_i\, p_j\, H\,(a_i - s)$ times, where r_i is the child's tokens-per-hour of CDS, p_j is the word's corpus probability, H is waking hours per month, and s is the global start time for accumulation. The child produces word j on the CDI with probability $\sigma(\log T_{ij} + \log \alpha_i - \psi_j)$ under a canonical Rasch/IRT logit link ($\sigma(x) = 1/(1 + e^{-x})$), where α_i is the child's learning-efficiency multiplier and ψ_j is the word's log-threshold. Every quantity lives on a log-token scale with $O(1)$ priors, so Stan is well-conditioned. The crucial move is that r_i's population distribution is pinned by external data (Sperry/HR/WF), so σ_α is identifiable and Question 4 has an answer. Questions 1–3 are posterior summaries of the baseline; planned variants sharpen them.

3 Quantities

Symbol	Meaning	Units
r_i	Child i's CDS input rate	tokens / hour
α_i	Child i's learning-efficiency multiplier	dimensionless
p_j	Word j's corpus probability (CHILDES)	dimensionless, $\sum_j p_j = 1$
H	Waking hours per month (fixed)	hours / month
a_i	Age at CDI assessment	months
s	Global start time (accumulation onset)	months
ψ_j	Log expected tokens of word j to acquisition	log-tokens
δ	Age rate-change exponent	dimensionless
λ_j	Word j's 2PL discrimination	dimensionless (> 0)
a_0	Reference age (fixed)	months
y_{ij}	Child i produces word j on CDI	$\{0, 1\}$

Fixed constants. $H = 12 \times 30.44 \approx 365$ hours/month (12 waking hours/day). $a_0 = 20$ months.

4 Generative model

4.1 Linear predictor

For child i, word j, age $a_i > s$:

$$\eta_{ij} = \lambda_j \cdot \Big[\log r_i + \log \alpha_i + \log p_j + \log H + (1 + \delta)\, \log\Big(\tfrac{a_i - s}{a_0}\Big) - \psi_j\Big]. \tag{1}$$

Here $\lambda_j \geq 0$ is a word-specific *discrimination* parameter (2PL). With $\lambda_j \equiv 1$ the model collapses to the Rasch form $\eta_{ij} = \log(r_i \alpha_i p_j H(a_i - s)) - \psi_j$ — the log ratio of tokens heard to tokens needed, with unit logit-per-log-token gain. With λ_j free, each word has its own gain; high λ_j produces a sharp transition near ψ_j, low λ_j a gradual one. Section **??** motivates the extension in terms of the accumulator and in terms of model fit.

4.2 Likelihood

Canonical Rasch/IRT logit link:

$$y_{ij} \mid \eta_{ij} \sim \text{Bernoulli}\big(\sigma(\eta_{ij})\big), \qquad \sigma(x) = \frac{1}{1 + e^{-x}}. \tag{2}$$

The logit link is the standard "log-linear IRT" choice and is numerically well-behaved in Stan. ψ_j is interpretable as the log-threshold in log-tokens on the logit scale (the scale factor is absorbed implicitly; $\exp(\psi_j)$ is proportional to, not exactly, the expected number of tokens to 50%-acquisition).

Edge case: when $a_i \leq s$, set $\mathbb{P}(y_{ij} = 1) = \varepsilon$ and exclude these cells from the log-age term (in the code this is implemented by flooring $a_i - s$ at 0.01 inside the log).

4.3 Priors

Child-level latent (collapsed). The likelihood depends on $\log r_i$ and $\log \alpha_i$ only through their sum, so we sample a single child-level latent

$$\xi_i \coloneqq \log r_i + \log \alpha_i \sim \mathcal{N}\big(\mu_r,\ \sigma_r^2 + \sigma_\alpha^2\big).$$

Here $\mu_r = \log 1198$ and $\sigma_r^2 = \mathrm{Var}(\log r)_{\text{Sperry/HR/WF}}$ are *pinned* from external data (this is the identifiability lever for RQ4), while σ_α is free:

$$\sigma_\alpha \sim \mathrm{HalfNormal}(0, 1). \tag{3}$$

The per-child $\log \alpha_i$ is a *derived* quantity with closed-form posterior conditional on ξ_i:

$$\log \alpha_i \mid \xi_i \ \sim\ \mathcal{N}\left(\frac{\sigma_\alpha^2}{\sigma_r^2 + \sigma_\alpha^2}\,(\xi_i - \mu_r),\ \frac{\sigma_r^2 \sigma_\alpha^2}{\sigma_r^2 + \sigma_\alpha^2}\right),$$

reported in generated quantities. (Rationale for this collapsed form is in §**??**: sampling both latents individually over-parameterizes the model and creates bad posterior geometry in HMC.)

Word thresholds by lexical class.

$$\psi_j \mid \mathrm{class}(j) = c \sim \mathcal{N}(\mu_c, \tau_c^2), \quad \mu_c \sim \mathcal{N}(8, 3^2), \quad \tau_c \sim \mathrm{HalfNormal}(0, 1). \tag{4}$$

Note on μ_c. The earlier value $\log 1000$ was a back-of-envelope centering. A more principled choice: tokens heard of a typical word by age 20 mo spans roughly e^5 (rare words) to e^{12} (function words), so $\mu_c \sim \mathcal{N}(8, 3^2)$ is weakly informative across the relevant range and dominated by the data.

Global start time (RQ2).

$$s \sim \mathcal{N}(4.5, 2^2) \text{ truncated to } [0, 15]. \tag{5}$$

One s for the whole model, per your decision. The prior is centered at 4.5 mo — M&P's comprehension estimate was ∼2 mo; production is likely later. s close to 0 would indicate accumulation begins near birth.

Age rate-change (RQ3).

$$\delta \sim \mathcal{N}(0, 0.5^2). \tag{6}$$

Word discrimination (2PL).

$$\log \lambda_j \sim \mathcal{N}(0, \sigma_\lambda^2), \quad \sigma_\lambda \sim \text{HalfNormal}(0, 1). \tag{7}$$

Setting $\sigma_\lambda \to 0$ (via a near-zero prior SD passed as data) pins every $\lambda_j = 1$ and recovers the Rasch model. This is the toggle used in code: one Stan file serves both Rasch and 2PL variants. Scale identifiability (the classic $\lambda_j \leftrightarrow 1/\lambda_j$, $\psi_j, \xi_i \leftrightarrow \lambda_j\psi_j, \lambda_j\xi_i$ ambiguity of 2PL) is resolved by our pinning of σ_r: the ξ scale is fixed externally, so ψ_j and λ_j are separately identified.

$\delta = 0$ is the pure cumulative null. $\delta > 0$ indicates older children accumulate faster — a formal test of the Bergelson comprehension-boost claim.

5 How each RQ is answered

RQ1 — frequency and word difficulty

The baseline has coefficient 1 on $\log p_j$ (the pure-accumulator assumption). Posterior samples $\hat{\psi}_j^{(s)}$ are residual word difficulty after frequency. We report:

- Posterior correlation $r(\hat{\psi}_j, \log p_j)$ and its 95% CrI.
- R^2 of $\hat{\psi}_j \sim \log p_j$ — what fraction of between-word variance in residual difficulty is explainable by frequency.
- Posterior class means $\hat{\mu}_c$ — does class structure persist after frequency is accounted for?

Planned variant (model B). Free global slope β on $\log p_j$; prior $\beta \sim \mathcal{N}(1, 0.5)$. Posterior on β directly tests the unit-coefficient assumption.

Planned variant (model C). Class-specific slopes β_c; answers whether frequency matters differently for nouns vs. verbs vs. closed class.

The three models form a progression (baseline $\subset$ B $\subset$ C). LOO-CV selects among them. Our headline RQ1 number is the pair (correlation in baseline, posterior on β in B) — one tells us *whether* frequency fully accounts for difficulty, the other *by how much* it under/over-weights.

RQ2 — start time

Single parameter s. Headline posterior: $\hat{s}$ with CrI. Close to 0 $\Rightarrow$ accumulation starts at birth. Close to 12+ $\Rightarrow$ delayed onset. The CDI is a production measure, so larger s than M&P's $\sim$2 months (comprehension) is expected.

RQ3 — acceleration

Single parameter δ. Headline posterior: $\hat{\delta}$ with CrI. Zero-excluded CrI $\Rightarrow$ rate change over age is needed.

RQ4 — input- vs. efficiency-driven variance

The linear predictor contains $\log r_i + \log \alpha_i$ as a sum, unidentifiable from Wordbank alone. We break the degeneracy by pinning σ_r^2 at its externally-measured value (from Sperry, Hart & Risley,

and Weisleder & Fernald pooled). The child-level latent $\xi_i = \log r_i + \log \alpha_i$ is then identified from the data, with implied distribution $\xi_i \sim \mathcal{N}(\mu_r, \sigma_r^2 + \sigma_\alpha^2)$. Subtracting the pinned σ_r^2 from the inferred $\mathrm{Var}(\xi_i)$ identifies σ_α^2, giving

$$\pi_\alpha \coloneqq \frac{\sigma_\alpha^2}{\sigma_r^2 + \sigma_\alpha^2} \tag{8}$$

as a well-defined derived posterior quantity: the proportion of child-level outcome variance attributable to learning efficiency rather than input rate. Its complement, $1 - \pi_\alpha$, is the input-driven fraction.

Why the collapsed parameterization matters practically. Sampling $\log r_i$ and $\log \alpha_i$ individually — each with its own non-centered $\mathcal{N}(0, 1)$ prior and a scale parameter — leaves the posterior with a flat ridge: any rotation of the pair that preserves ξ_i has the same likelihood. Only the priors break the degeneracy, at the cost of 2I parameters where I are identified. In HMC this manifests as chains getting stuck in bad regions (we observed one chain running 28+ minutes on a sim the others cleared in 3). Sampling ξ_i directly, with σ_α entering only through $\sigma_{\xi_i}^2 = \sigma_r^2 + \sigma_\alpha^2$, eliminates this geometry entirely. Parameter recovery on a 250-child $\times$ 150-word synthetic dataset: π_α posterior 0.62 [0.50, 0.72] against true 0.61.

Sensitivity analysis. Refit with $\sigma_r^2 \in \{0.5, 1, 2\} \cdot \sigma_r^{2,\mathrm{obs}}$. Stability of π_α across these indicates how much the claim depends on trusting the external estimate.

6 Why 2PL: one-token-one-drop is too strong

The Rasch version of (**??**) (i.e. with $\lambda_j \equiv 1$) implies every word has the same logit-per-log-token gain. For a fixed word, the predicted slope of $\mathrm{logit}\, p_{ij}$ with respect to age is $(1 + \delta)/(a_i - s)$ — on the order of 0.08 logits per month at typical observed ages. Real CDI words transition from $\sim$5% to $\sim$95% acquired over roughly six months (about 6 logits), which is $\sim$12$\times$ faster than the Rasch slope can produce.

When we fit Rasch to Wordbank (500 children $\times$ 200 items), the posterior finds this mismatch in a predictable way: it cranks every available steepening lever — δ rises to $\sim$ 2.3, s pushes to $\sim$ 12 months, and crucially σ_α inflates to $\sim$ 2 — implying children differ by $\sim$ 50$\times$ in learning efficiency, and $\pi_\alpha \approx 0.93$. This pattern is stable across the s/δ identifiability, suggesting misspecification rather than genuine individual-difference variance.

The 2PL fix. Adding word-specific discrimination λ_j gives each word its own slope around threshold: the per-word logit derivative becomes $\lambda_j(1 + \delta)/(a_i - s)$. Words whose referents are highly diagnostic of their meaning ("ball" when you see a ball; concrete nouns generally) can have large λ_j and transition sharply; words whose meaning is more distributional or context-dependent (function words, abstracts) have small λ_j and gradual transitions. This is the standard two-parameter logistic IRT extension and is strongly preferred over 1PL for CDI data in prior work (Kachergis, Marchman, Dale, Mankewitz, & Frank, 2022, *JSLHR*).

Three accumulator interpretations of λ_j (all observationally equivalent from CDI alone).

1. **Information per token.** λ_j is the per-token shift in the child's certainty about word j. High λ_j: each exposure is diagnostic.
2. **Sharpness of the cognitive threshold.** λ_j is the inverse width of the band over which evidence must accumulate for acquisition to occur.
3. **ELI-to-token ratio (bridge to Mollica & Piantadosi).** If only a fraction of heard tokens are effective learning instances and the rest are background noise, λ_j is proportional to that fraction for word j.

Falsification checks. The hierarchical prior on $\log \lambda_j$ (pooled via σ_λ) regularizes: one unusual word can't take over. Two further checks protect against freely flexible fit-to-noise:

1. σ_α should *decrease substantially* under 2PL. If the Rasch inflation was misspecification dressed as person variance, 2PL should absorb it into λ_j and drop σ_α toward a more reasonable value (e.g. $\sim$0.7), sharpening our answer to RQ4.
2. λ_j should correlate with measurable word properties: higher for concrete nouns, lower for function words and abstract adjectives. If λ_j is uncorrelated with any external measurement, we have over-fit rather than learned.

What this changes for the RQs. RQ1 (frequency-vs-difficulty) unchanged: we still report the posterior correlation of ψ_j with $\log p_j$. RQ2 (start time), RQ3 (δ) unchanged at the structural level, but the extreme posteriors observed under Rasch should moderate. RQ4 (variance decomposition) sharpens substantially: π_α becomes the proportion of child variance after item-level discrimination has been accounted for, which is the quantity we really want.

7 Assumptions, listed

1. **Pure accumulation.** Exposure accumulates linearly at a per-token rate independent of prior knowledge. No leveraged learning (Mitchell & McMurray 2009 showed leverage cannot *create* acceleration that is not already present).
2. **Per-word evidence gain λ_j.** In 2PL mode each token contributes λ_j units of logit-scale evidence; $\lambda_j = 1$ reduces to the Rasch "one token, one drop" rule. Per-child gain is *not* modeled: all children receive the same λ_j for a given word. This is relaxable by adding a random slope, which we defer.
3. **Logit link with fixed scale.** Canonical Rasch/2PL IRT. Equivalent (up to a ~ 1.7 scale factor) to a probit model; approximates the Gamma cumulative of Mollica & Piantadosi for typical parameter values.
4. **Cross-sectional data.** Each child observed once.
5. **CDS only.** Overheard speech ignored in v1.
6. **Same input distribution as reference studies.** Our CDI sample has the Sperry/HR/WF distribution of r_i; SES moderators not modeled in v1.
7. **Global s.** One start time for all words. Per-class or per-word s is a planned extension.

8. **CHILDES frequencies as truth.** Sampling noise in p_j ignored; defensible for common words.
9. **Lexical class only via thresholds.** Class enters through the hierarchical prior on ψ_j; class-specific frequency slopes or start times are variants, not baseline.

8 Sampling strategy: children and items

The data we fit is large enough that we always work on a subsample. The strategy is the same across pipelines (Wordbank longitudinal, BabyView, future Seedlings) so results are directly comparable.

Children.

1. Filter to children with ≥ 2 admins (longitudinal coverage).
2. Compute each child's *median admin age* as their representative age.
3. Bin children into $N_{\text{age}} = 4$ age quantile bins.
4. Within each bin, take the top kids by number of admins, then by age span. (Tie-breaking on richer longitudinal data.)
5. Top up uniformly at random if any bin is short (rare).

This replaces an earlier, biased "top-by-priority" rule that just maximized $n_{\text{admin}} + \text{span}/5$; that rule overweighted data-rich kids in narrow age windows. The age-stratified rule guarantees coverage across the whole 16–30 mo developmental window.

Items.

1. Filter to items with non-zero CHILDES probability p_j.
2. For each lexical class, bin items into $N_{\text{diff}} = 4$ difficulty quartiles based on $\log p_j$.
3. Sample uniformly within each (class, difficulty-quartile) cell — 16 cells for 4 classes $\times$ 4 quartiles.
4. Top up at random if any cell is short.

The earlier item-sampling rule stratified only on lexical class, which oversampled high-frequency words (those are dense in the CHILDES distribution). The class-by-difficulty grid ensures we have hard, medium, and easy words within *each* class, which sharpens the identification of ψ_j and β_c when they enter as parameters.

Pipeline-shared item pool. All Wordbank-based pipelines (cross-sectional and longitudinal, English and Norwegian) draw from the *union of WG and WS forms*, mirroring the BabyView treatment. Each (child, age, form) is its own admin; a child who took both WG and WS at different ages contributes both. This extends English age coverage from the WS-only 16–30 mo to 8–30 mo and matches the cross-pipeline data definition.

Per-pipeline application.

- **Wordbank longitudinal (English / Norwegian).** 200 children $\times$ 200 items per dataset, drawn via the stratified sampling above. English: median 3 admins, 14 mo span; Norwegian: median 8 admins, 14 mo span (Norwegian has the much denser Kristoffersen contributor).
- **Wordbank cross-sectional (English).** 500 children, treated as the most recent (child, form) admin from `long_items.rds`. Stratified the same way as the longitudinal pipeline.
- **BabyView.** 20 longitudinal English subjects (the entire available pool) $\times$ 200 items, with input rate observed via 5 688 videos. No child-side subsampling: we use everyone.

9 What we deliver

1. **RQ-addressed posteriors**: π_α, s, δ, and $r(\hat{\psi}_j, \log p_j)$ all with credible intervals.
2. **Per-word** ψ_j: absolute log-token thresholds for all 680 CDI items.
3. **Per-child** $\log \alpha_i$: child-level efficiency estimates.
4. **Counterfactual simulator**: given hypothetical (r, α, a), predicted vocabulary size with CrI.
5. **Planned model comparisons**: LOO-CV across the baseline plus variants for each RQ (see §8).

10 Lean baseline + opt-in variants

After running through several rounds of fits with all the bells and whistles attached by default, we settled on a deliberately *lean* baseline and a small set of variants that opt in to extra parameters explicitly.

Lean baseline (the default). Includes:

- Per-child latent ξ_i (intercept; carries the σ_α / π_α decomposition via the externally-pinned σ_r).
- Per-word ψ_j with class-hierarchical prior.
- Frequency $\log p_j$ entering the linear predictor with unit coefficient (kept separate so RQ1 can be answered as $r(\psi_j, \log p_j)$).
- Population acceleration exponent δ, free.
- Class means μ_c and SDs τ_c.

Pinned at zero by default. Both for parsimony and because they are weakly identified in cross-sectional data:

- Start time $s = 0$ (pinned via near-zero prior SD).
- 2PL discrimination $\lambda_j \equiv 1$.
- Per-child slope deviations $\zeta_i \equiv 0$.

Variants opt in.

Variant	Adds	Question / use
`slopes`	ζ_i, σ_ζ	Do children vary in growth rate? Default for longitudinal.
`2pl`	λ_j, σ_λ	Do words vary in acquisition sharpness?
`2pl_slopes`	both	Comparison; defends dropping 2PL when it adds little.
`free_s`	free s	RQ2 robustness check.
`fix_delta`	$\delta = 0$	Ablation: no acceleration.

The running log of fits and what each revealed is in `notes/experiments.md`.

Why we changed defaults. The earlier defaults included free s, 2PL, and slopes. Three findings drove the simplification: (a) 2PL never measurably improved population-level fit even though $\sigma_\lambda \approx 0.27$ is non-zero (item discrimination exists but the population averages it out); (b) start time s was poorly identified, hitting prior bounds in cross-sectional fits and trading off strongly with δ; (c) per-child slopes σ_ζ were unidentifiable from cross-sectional data (needed longitudinal). The lean default is the smallest model that addresses every RQ; the variants exist to test specific claims.

11 Datasets and observation channels

The variant grammar (§ Lean baseline) toggles *model components*. Independent of that, each dataset comes with its own *observation channels* that determine which auxiliary parameters are appropriate. We separate the two levers in code so they can't be mixed by accident.

Lever 1 — variant grammar (per-fit toggles). Tight-prior switches on σ_λ, σ_ζ, s, δ. Implemented in `variant_hyperpriors()` (`model/R/helpers.R`); identical across pipelines (the `long_`, `io_` prefixes are stripped inside).

Lever 2 — dataset-level design choices (per-bundle constants). Properties inherent to the observation channel of a given dataset. Baked into `stan_data` by the `prepare_*.R` script; not a choice a researcher makes per fit. Two examples currently:

- *Reactivity inflation β_{react}.* Per-child observed input rate $\log r_{iv}^{\text{obs}}$ for video v of child i is $\log r_i + \beta_{\text{react}} + \varepsilon_{iv}$. The bias β_{react} absorbs systematic differences between observed input and "true" input.
  - **BabyView** (head-mounted video, on-camera observer effect): $\beta_{\text{react}} \sim \mathcal{N}(0.4, 0.4)$ — active, with the prior centered around $\sim$1.5x inflation.
  - **SEEDLingS** (LENA all-day audio, no on-camera observer): $\beta_{\text{react}} \sim \mathcal{N}(0, 0.001)$ — pinned at zero. Passive-recording channel, so no inflation.
- *Per-recording weighting.* Currently uniform; planned hook in `stan_data$log_r_obs_weight` for downweighting short or contaminated recordings. Off by default.

Stan model files form a partial order. Each file *adds* components on top of the previous, all toggleable via tight priors:

File	Adds	Used by
`log_irt.stan`	cross-sectional baseline ξ_i	Wordbank cross-sectional fits
`log_irt_long.stan`	admin→child indexing, ζ_i, bivariate prior	Wordbank longitudinal fits
`log_irt_io.stan`	per-recording $\log r^{\text{obs}}$ measurement layer, β_{react}	BabyView, SEEDLingS
`log_irt_proc.stan` *(planned)*	noisy-α observation from LWL RT/accuracy	Peekbank-linked Stanford datasets

The result is a small partial order of model files, not a combinatorial explosion. Adding a new dataset usually means writing a `prepare_<dataset>.R` script that reuses an existing Stan file with appropriate dataset-level constants; only when a genuinely new *observation channel* appears do we add a Stan file.